

DAY 05 – ETHERNET LAN SWITCHING FUNDAMENTALS

Networking Assignment Report

Student Name	Hamza Khan
Assignment Code	DAY05-ETHERNET-LANSWITCHING
Total Marks	20
Operating System	Ubuntu Linux

Ubuntu Linux Note: The assignment specifies Windows commands such as getmac and ipconfig /all. This practical was performed on Ubuntu Linux, so the Linux equivalents ip link show, ifconfig -a, and ip neigh were used in place of getmac, ipconfig /all, and arp -a respectively.

ETHERNET LAN SWITCHING – THEORY

1. What is Ethernet?

Ethernet is a prevalent technology for local area networks (LANs) that provides a standardized means of communication over networks. Ethernet is used extensively in home networking, office buildings, schools, data centers, and enterprise networks.

2. What is an Ethernet frame?

An Ethernet frame is a packet of data transferred and communicated on an Ethernet network. An Ethernet frame comprises a Destination MAC Address, a Source MAC Address, a Type/Length indicator for the payload, the Payload itself, and a Frame Check Sequence (FCS) utilized for error-checking purposes.

3. What is a MAC address?

A MAC (Media Access Control) address is a unique hardware address at Layer 2 associated with a network interface, e.g., 00:1A:2B:3C:4D:5E. Unlike an IP address, which identifies a device logically at Layer 3, a MAC address is used to uniquely identify an interface at Layer 2.

4. What is the purpose of the Source MAC address?

The source MAC address identifies the interface from which the frame originated. A switch utilizes this field to learn about the interfaces of connected devices.

5. What is the purpose of the Destination MAC address?

The destination MAC address field specifies the intended receiver of the frame. A switch uses this field to determine which interface to send out the frame.

6. What is a MAC Address Table?

A MAC Address Table is a table on a switch that stores a database of learned and discovered MAC addresses and the associated interfaces or ports. All Layer 2 frame forwarding decisions made by a switch are performed using the information contained in this table.

7. How does a switch learn MAC addresses?

When a device sends a frame to a switch, the switch examines the source MAC address. The switch then saves this address in its MAC Address Table along with the interface on which the frame was received to identify the location of each of its directly connected interfaces. For example, for PC-A, a switch will know that its MAC address is located on Port 1 whenever it receives any frame sent by PC-A.

8. What is unknown unicast flooding?

Unknown unicast flooding is a process that occurs when a switch receives a frame whose destination MAC address was not found in its MAC Address Table. In such a scenario, the switch cannot determine the specific port to forward the frame and, consequently, floods the frame out of all ports within the same Layer 2 broadcast domain, excluding the port from which the frame was received.

9. What is known unicast forwarding?

In known unicast frame forwarding, a switch consults its MAC Address Table to determine the appropriate port for the destination MAC address specified in the frame and only forwards the frame out of that particular port instead of all ports.

10. Compare MAC addresses and IP addresses.

A MAC address is used at Layer 2 for local interface identification (e.g., 00:1A:2B:3C:4D:5E), while an IP address is used at Layer 3 for identifying devices and routing packets across different networks (e.g., 192.168.1.10).

11. Why does Ethernet switching matter in cybersecurity?

Cybersecurity professionals conduct extensive analysis of MAC addresses, Ethernet frames, local traffic, switches, and Address Resolution Protocol (ARP) tables in order to perform local traffic analysis, network segmentation, or conceptualize Layer 2-based attacks such as MAC flooding or ARP spoofing. This knowledge will form the basis for future courses involving the utilization of tools such as Wireshark and Nmap for cybersecurity applications, networking, and incident response and management.

PRACTICAL — IDENTIFYING MAC ADDRESSES AND LOCAL NETWORK INFORMATION

The assignment requests `getmac`, `ipconfig /all`, and `arp -a`. Since this practical was performed on Ubuntu Linux, the equivalent commands `ip link show`, `ifconfig -a` and `ip neigh` were used instead.

12. Recorded adapter and IP information (Task 1)

Field	Value
Adapter name	wlp0s20f3 (Wi-Fi)
MAC address	c8:09:a8:e6:1e:f5
IPv4 address	172.20.10.3
Subnet mask	255.255.255.240
Broadcast address	172.20.10.15
Default gateway	172.20.10.1

Connection type	Wi-Fi
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Other network interfaces present on the system but not carrying traffic at capture time:

Adapter	MAC address	Status
enp0s31f6 (Ethernet)	e8:d8:d1:ec:91:9e	DOWN, no cable connected
wlx00e02691d31b (USB Wi-Fi)	00:e0:26:91:d3:1b	DOWN, not in use
lo (Loopback)	N/A	UP, 127.0.0.1

13. ARP / neighbour cache entry (Task 2)

Output of ip neigh (Linux equivalent of arp -a) showed the router's IP-to-MAC mapping:

IP address	MAC address	State
172.20.10.1 (gateway)	ea:5f:02:2f:76:64	REACHABLE, router

14. Matching practice (Task 3)

Concept	Layer
MAC address	Layer 2
IP address	Layer 3
Switch (forwarding)	Layer 2
Router (routing)	Layer 3
Ethernet frame	Layer 2

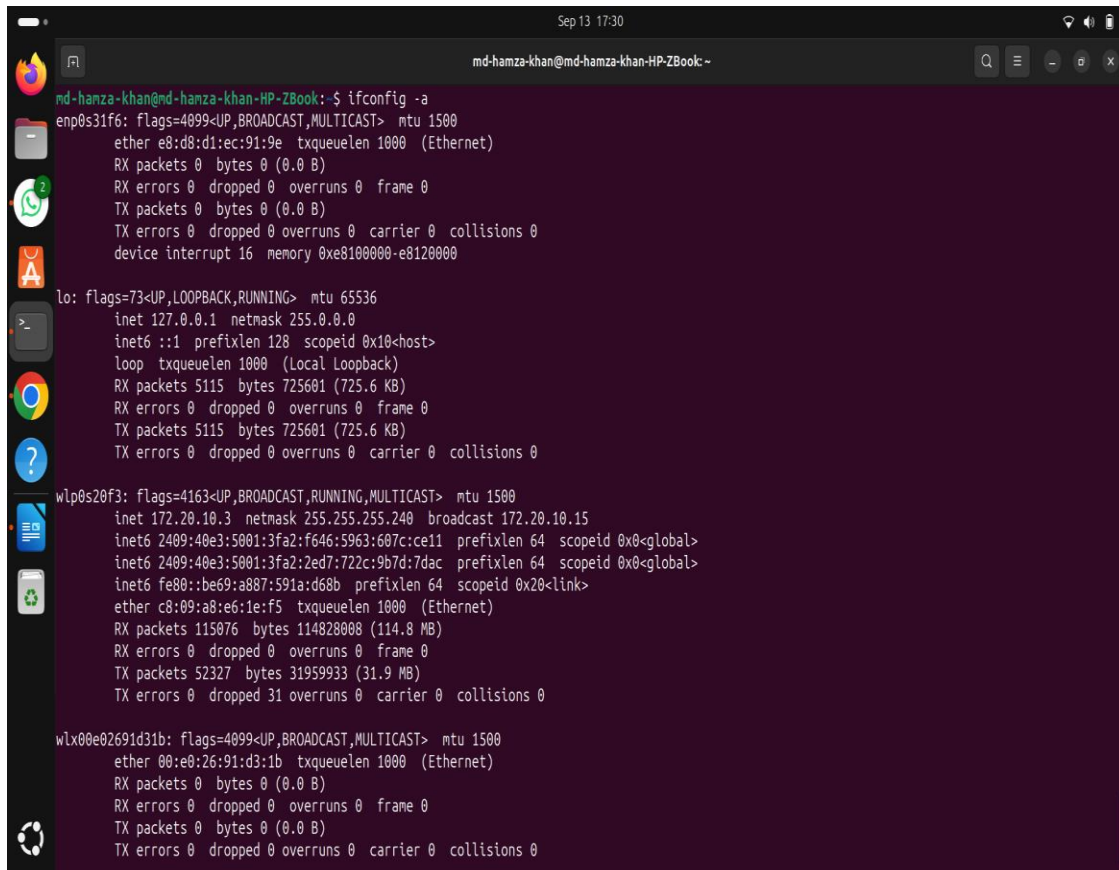
SCREENSHOTS

```

md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip neigh
172.20.10.1 dev wlp0s20f3 lladdr ea:5f:02:2f:76:64 REACHABLE
2409:40e3:5001:3fa2:dcd6:da0c:ef05:9e0d dev wlp0s20f3 lladdr ea:5f:02:2f:76:64 router REACHABLE
fe80::e85f:2ff:fe2f:7664 dev wlp0s20f3 lladdr ea:5f:02:2f:76:64 router REACHABLE
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
    link/ether e8:d8:d1:ec:91:9e brd ff:ff:ff:ff:ff:ff
3: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT group default qlen 1000
    link/ether c8:09:a8:e6:1e:f5 brd ff:ff:ff:ff:ff:ff
5: wlx00e02691d31b: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DORMANT group default qlen 1000
    link/ether 00:e0:26:91:d3:1b brd ff:ff:ff:ff:ff:ff

```

Figure 1 – ip neigh and ip link show



```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ifconfig -a
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether e8:d8:d1:ec:91:9e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xe8100000-e8120000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 5115 bytes 725601 (725.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5115 bytes 725601 (725.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp0s20f3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.20.10.3 netmask 255.255.255.240 broadcast 172.20.10.15
    inet6 2409:40e3:5001:3fa2:f646:5963:607c:ce11 prefixlen 64 scopeid 0x0<global>
    inet6 2409:40e3:5001:3fa2:2ed7:722c:9b7d:7dac prefixlen 64 scopeid 0x0<global>
    inet6 fe80::be69:a887:591a:d68b prefixlen 64 scopeid 0x20<link>
    ether c8:09:a8:e6:1e:f5 txqueuelen 1000 (Ethernet)
    RX packets 115076 bytes 114828008 (114.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 52327 bytes 31959933 (31.9 MB)
    TX errors 0 dropped 31 overruns 0 carrier 0 collisions 0

wlx00e02691d31b: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 00:e0:26:91:d3:1b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 2 – ifconfig -a

COMMANDS USED

ip link show

ifconfig -a

ip neigh

Author: Hamza Khan

Platform: Ubuntu Linux